Applied Acoustics 241 (2026) 110960

Contents lists available at ScienceDirect

Applied Acoustics

journal homepage: www.elsevier.com/locate/apacoust

Reference-free singing voice MOS prediction via multi-feature fusion, with integrated feature analysis

Peng Bai [a,c], Yue Zhou [a,c], Ke Gu [b,c], Meizhen Zheng [a,c], Linshujie Zheng [d], Yidong Chen [a,b,c], Xiaodong Shi [a,b,c,*]

[a] Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen, 361005, Fujian, China
[b] Institute of Artificial Intelligence, Xiamen University, Xiamen, 361005, Fujian, China
[c] Key Laboratory of Digital Protection and Intelligent Processing of Intangible Cultural Heritage of Fujian and Taiwan (Xiamen University), Ministry of Culture and Tourism, Xiamen, 361005, Fujian, China
[d] Tsai Chi-Kun Academy of Music, Minjiang University, Fuzhou, 350108, Fujian, China

ARTICLE INFO

Keywords:
Mean opinion score prediction
Singing voice assessment
Gezi opera singing
Self-supervised learning model

ABSTRACT

The singing voice Mean Opinion Score (MOS) prediction model is crucial for assessing vocal quality and is increasingly adopted as an alternative to manual evaluation. However, current research faces three key challenges. First, publicly available datasets are scarce. Second, existing models have not sufficiently leveraged relevant features. Third, single-score assessments provide limited insights into specific weaknesses, making it difficult to improve vocal quality without fine-grained analysis. To address these challenges, we first construct a publicly accessible dataset for predicting the MOS of Hokkien Gezi Opera singing. The dataset integrates diverse sources, maintains a well-balanced score distribution, and includes true ratings from six professionals. Second, we propose a novel framework comprising a reference-free MOS prediction module and an integrated feature analysis module. The MOS prediction module, built on a speech self-supervised learning model, employs multi-feature fusion to comprehensively predict singing MOS from the perspectives of initial perception, pronunciation clarity, timbre, pitch, and emotion. Notably, this is the first model to incorporate pronunciation clarity and emotion as evaluation feature. Simultaneously, the integrated feature analysis module provides reference-based auxiliary feedback across four aspects: pronunciation clarity, timbre, pitch, and emotion. Experiments on our constructed Hokkien Gezi Opera dataset, along with publicly available Chinese-singing and English-speaking datasets, demonstrate that our Multi-Feature Fusion MOS (MFFMOS) prediction model achieves state-of-the-art performance across four key metrics: Mean Square Error (MSE), Linear Correlation Coefficient (LCC), Spearman Rank Correlation Coefficient (SRCC), and Kendall Tau Rank Correlation Coefficient (KTAU). Furthermore, user study involving fifteen students in the singing education context confirms that our integrated feature analysis module is practical and helps them understand their weaknesses in specific aspects of their singing. The strong predictive performance and domain adaptability of the MOS prediction module, combined with the practicality of the feature analysis module, establish our framework as a compelling alternative to manual evaluation.

1. Introduction

Singing voice Mean Opinion Score (MOS) prediction is a task that involves evaluating the quality of singing voice audio, whether sung by humans or synthesized by models, through the prediction of a score representing its perceived quality [1]. In this research field, “singing voice” refers specifically to the pure singing voice, without any accompanying music. This task has substantial application value in the automated assessment of singing quality, with particular relevance to online music education, karaoke platforms, and the automated subjective evaluation of Singing Voice Synthesis (SVS) models. SVS refers to synthesizing singing voices from lyrics and music scores through machine learning models [2]. Currently, the evaluation in the SVS field primarily relies on manual rating. The subjective evaluation results of SVS is often difficult to replicate across studies, as the varying conditions and backgrounds of different raters make it challenging to fairly compare SVS models ac-

* Corresponding author at: Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen, 361005, Fujian, China.
E-mail addresses: baipeng@stu.xmu.edu.cn (P. Bai), zhouyue1@stu.xmu.edu.cn (Y. Zhou), gukexmu@stu.xmu.edu.cn (K. Gu), midon@stu.xmu.edu.cn (M. Zheng), zhenglinshujie@mju.edu.cn (L. Zheng), ydchen@xmu.edu.cn (Y. Chen), mandel@xmu.edu.cn (X. Shi).

https://doi.org/10.1016/j.apacoust.2025.110960
Received 5 April 2025; Received in revised form 23 June 2025; Accepted 16 July 2025

Available online 13 August 2025
0003-682X/

cording to a unified standard [3]. Subjective human evaluation is also both time-consuming and costly. When a large number of evaluation samples are involved, evaluators' performance may fluctuate, resulting in a gradual decline in the accuracy of their assessments over time. As a result, researchers have begun exploring the automatic MOS prediction for singing voice, whether sung by humans or synthesized by SVS models.

In terms of data, compared to the speech MOS prediction dataset, the publicly available dataset for singing voice MOS prediction is relatively scarce, and it is primarily focused on Chinese and English pop songs. Due to copyright and privacy issues associated with some singing recordings, researchers are unable to release data publicly [4–6]. This creates significant difficulties for research in this field. Regarding the need for reference audio, singing voice MOS prediction models can be categorized into two types: reference-based and reference-free [6]. An reference-based MOS prediction model requires a reference audio, which is often unavailable in real-world scenarios. In contrast, a reference-free model does not require reference audio, making it more practical and valuable. Study [7] has shown that music experts can consistently evaluate new melodies or vocal performances they have never encountered before, drawing on their accumulated musical knowledge and intuitive judgment. This demonstrates the feasibility and practical value of reference-free singing MOS prediction. In terms of model design, current research on singing voice MOS prediction primarily follows three approaches: leveraging only spectrogram features [4,8], utilizing only acoustic features such as pitch, rhythm, and timbre [9–11], and adopting hybrid methods that integrate both spectrogram and acoustic features [1,5,6,12].

We note that the professional music vocal study [13] identifies key aspect-level terms in singing voice assessment, namely timbre, pitch, text, and expression, which are the four aspects considered by experts during their manual evaluations. However, the exploration of acoustic features in existing MOS prediction models remains somewhat limited, as two crucial elements in singing, pronunciation clarity (referred to as "text" by vocal experts) and emotion (referred to as "expression" by vocal experts), have yet to be fully incorporated. Furthermore, current studies focus primarily on predicting scores without offering a detailed analysis. This approach provides limited insights into specific weaknesses, hindering efforts to improve vocal quality. Based on the above analysis and comparison, it is evident that research on predicting the MOS of singing voices faces three key challenges: first, the lack of high-quality and publicly available datasets; second, existing models not fully utilizing relevant features; and third, the lack of analysis in single-score assessments makes it difficult to identify specific weaknesses, hindering efforts to enhance vocal quality without a more detailed analysis.

To address the first challenge, we expand the research scope and construct a high-quality Chinese Hokkien **G**ezi **O**pera **S**inging voice **MOS** prediction dataset, **GOSMOS**, which is the first publicly available MOS prediction dataset for Chinese opera in the world. Chinese opera is a treasure of traditional Chinese culture, and Gezi Opera, a unique form of Chinese opera, is an intangible cultural heritage of China.[1] The dataset integrates a wide range of diverse sources, including recordings from professional performers as well as multiple SVS-based audio samples, ensuring it can effectively support the evaluation of both human and SVS models. Additionally, the dataset includes ratings from six professional practitioners, each contributing their expertise in Hokkien Gezi Opera to provide accurate and reliable evaluations of the singing voices. It also ensures a well-balanced distribution of scores across various levels, further enhancing the dataset's reliability and utility.

To address the second and third challenges, and in contrast to existing models that only predict MOS, we propose a novel framework that includes a reference-free MOS prediction module and an integrated feature analysis module. Since the pre-trained speech Self-Supervised Learning (SSL) model has been proven to effectively capture intricate speech patterns and enhance performance in speech MOS prediction tasks [1,14], we propose a **M**ulti-**F**eature **F**usion **MOS** (**MFFMOS**) model based on SSL model that provides a comprehensive evaluation of the singing voice from five perspectives: initial perception, timbre, pitch, pronunciation clarity, and emotion. To the best of our knowledge, this is the first study in this field to explore pronunciation clarity and emotion in singing voices. The integrated feature analysis module provides reference-based auxiliary feedback across four aspects: pronunciation clarity, timbre, pitch, and emotion.

Our contributions can be summarized as follows:

- We construct GOSMOS, the world's first publicly available MOS prediction dataset for Chinese Hokkien Gezi Opera. This dataset is sourced from diverse origins, ensuring broad coverage, and maintains a well-balanced score distribution for reliable evaluation.
- We propose a novel framework that includes a reference-free Multi-Feature Fusion MOS (MFFMOS) model and an integrated feature analysis module. The MFFMOS model based on an SSL model evaluates singing voices across five dimensions: initial perception, timbre, pitch, pronunciation clarity, and emotion. Meanwhile, the integrated feature analysis module offers reference-based feedback on pronunciation clarity, timbre, pitch, and emotion, enabling targeted vocal improvement.
- Through extensive experiments on our constructed Hokkien Gezi Opera dataset, as well as publicly available Chinese-singing and English-speaking datasets, we demonstrate that the MFFMOS model achieves outstanding performance across four key metrics (MSE, LCC, SRCC, KTAU) at both the utterance level and system level. These results highlight the model's effectiveness and domain adaptability, and also demonstrate that our constructed dataset is well-suited for the MOS prediction task. Additionally, a user study involving 15 students further validates the practical utility of the integrated feature analysis module in the context of singing education.

The rest of this paper is organized as follows: Section 2 introduces related works on singing voice assessment and speech MOS prediction. Section 3 introduces the Gezi Opera singing voice MOS prediction dataset we have constructed. Section 4 presents our proposed framework. Section 5 describes our experimental datasets, comparison models, training details and evaluation metrics. Section 6 analyzes and discusses the experimental results. Section 7 provides a conclusion and outlook for future work.

2. Related works

2.1. Singing voice assessment

According to the presence or absence of reference audio, singing voice assessment can be divided into two categories: reference-based and reference-free. In terms of research about reference-based, Tsai and Lee [11] exploited various acoustic features such as pitch, volume, and rhythm to automatically evaluate karaoke singing. Molina et al. [15] proposed a universal method for automatic singing evaluation of basic singing skills. This method obtains intonation, rhythm, and overall score by measuring the similarity between singing and original singing. In addition to utilizing acoustic features, Gupta et al. [16] adopted the idea of PESQ and proposed various perceptual related features to evalu-

[1] https://www.ihchina.cn/art/detail/id/13290.html.

ate singing quality, which is called the Perceived Evaluation of Singing Quality (PESnQ) score.

In terms of research on reference-free, singing voice assessment primarily focuses on the task of singing voice MOS prediction. Current research on singing voice MOS prediction primarily follows three approaches: leveraging only spectrogram features [4,8], utilizing only acoustic features such as pitch, rhythm, and timbre [9–11], and adopting hybrid methods that integrate both spectrogram and acoustic features [1,5,6,12]. In the first approach, Zhang et al. [4] proposed a novel bidensenet to extract effective features from the singing spectrogram to classify the singing. Tan [8] proposed a relation network that enables the model to learn the differences between different singing. They also constructed a triplet network based on relation networks to achieve the goal of rating singing performances through metric learning. In the second approach, Nakano et al. [9] proposed earlier the use of pitch interval accuracy and vibrato features, which is effective for automatically evaluating singing skills without music score information. Gupta et al. [10] proposed a framework to evaluate the singing quality of a large number of singers without any standard reference. They define absolute measurements of musical attributes using pitch histograms and relative measurements based on inter-singer statistics. In the third approach, Huang et al. [12] designed a Convolutional Recurrent Neural Network (CRNN) for automatic singing quality evaluation and compared three spectrogram features: mel spectrogram, Constant-Q Transform (CQT) spectrogram, and chromogram, as network inputs, confirming that the CQT spectrogram performed the best. Ju et al. [6] proposed a more general automatic singing skill evaluation model suitable for solo and accompaniment singing. They studied timbre features, attention mechanisms, and the addition of dense layers to further improve performance based on CRNN. Sun et al. [5] proposed the TG-Critic model to explore the multi-scale backbone network of CNN with input of CQT. In addition to CQT, TG-Critic also uses timbre representation, on the basis of CRNN model, Chan et al. [1] fused the features of pre-trained speech SSL models and achieved better results. In terms of dataset construction, Zhang et al. [4] constructed an automatic singing evaluation dataset named SIE-22k. Ju et al. [6] constructed a 10K singing dataset. Sun et al. [5] constructed a YJ-16K dataset. Unfortunately, the datasets SIE-22k, 10K singing, and YJ-16K are not publicly available.

We have constructed and publicly released a Gezi Opera singing voice MOS prediction dataset for research purposes. Building upon existing acoustic features (pitch and timbre), our research utilizes SSL representations and incorporates pronunciation clarity and emotion features to enable a more comprehensive evaluation. In addition, our framework also provides feature-level analytical feedback to help users improve their singing quality.

2.2. Speech MOS prediction

The speech MOS prediction aims to save the cost of subjective evaluation in speech synthesis or voice conversion tasks. Speech MOS prediction can be divided into two categories: reference-based and reference-free [17]. The reference-based MOS prediction model takes an audio to be evaluated and a high-quality reference audio as input, and the model learns the differences between these two signals to predict the MOS [18]. The reference-free MOS prediction model only receives the audio to be evaluated as input, and the model analyzes the features of the audio signal itself to predict the MOS. Perceptual Evaluation of Speech Quality (PESQ) [19], ViSQOL [20], ViSQOL v3 [21], are among the widely utilize reference-based metrics for evaluating speech quality. The representative reference-free works for speech MOS prediction include MOSNet [22], MBNet [23], LDNet [24], NISQA [25] and MSQAT [26]. Most of these reference-free models utilize Convolutional Neural Network (CNN) or Recurrent Neural Network (RNN) to analyze speech spectrogram. In addition, there are also studies [14,27] that directly evaluate speech using pre-trained speech SSL models. This type of model, due to being pre-trained on a large amount of speech data, usually performs better than traditional neural networks. Due to the outstanding performance of pre-trained speech SSL models, we integrate their features with singing acoustic features to predict MOS.

3. GOSMOS dataset

3.1. Data collection

The audio data comes from three sources: first, recordings by professional singers; second, audio generated by singing voice synthesis models; and third, low-quality audio manually created from the first two types of audio. As for the first part of the audio, we invite five professional Gezi Opera singers to record singing audio in a quiet environment. The singing audio recordings feature three female singers and two male singers, all performing precisely in accordance with the music scores. Each female singer contributes 1 hour of audio, one male singer contributes 1 hour, and the other male singer contributes 0.5 hours. For the synthesized audio in the second part, we first select four SVS models. The field of Gezi Opera SVS has witnessed significant advancements, as demonstrated by specialized frameworks such as FT-GAN [3] and SVSELM [2], which are specifically designed for this traditional art form. We also use two classic SVS models: FFTSinger [28] and DiffSinger [29] to synthesize audios. Additionally, we employ two methods to generate low-quality SVS samples. The first involves synthesizing audio using a singing voice synthesis model that has not yet fully converged during training, resulting in severe noise. The second involves modifying the phoneme set during the inference stage, leading to synthesized audio with unclear and inaccurate pronunciation. The data in the third part is derived from manually degraded recordings and high-quality SVS-based audios to construct low-quality audio samples. We employ three approaches: slightly adjusting the pitch in specific phoneme regions, slightly adjusting the energy in specific phoneme regions, and significantly altering both pitch and energy in specific phoneme regions.

3.2. Human rating

In the human rating process, we invite six Hokkien Gezi Opera professionals to rate each singing voice audio file. The six professionals possess a certain level of appreciation for Gezi Opera singing. Prior to rating, they follow a blind listening procedure to ensure that they are not influenced by any preconceived biases. The evaluators do not receive any background information about the singers or any other details of the audio (such as the singer's gender, age, or whether the audio is based on SVS model). To prevent auditory fatigue from listening to multiple singing clips consecutively, each evaluator is limited to a certain number of audio files per session, with breaks provided to maintain the accuracy and fairness of the scores. The evaluators give a score to each audio clip, using a 1 to 5 scale with intervals of 0.5. Specifically, a score of 1 indicates "poor", 2 indicates "fair", 3 indicates "average", 4 indicates "good", and 5 indicates "excellent". The score distribution of the six evaluators is presented in Fig. 1(a) to Fig. 1(f). After the evaluation, the ground truth MOS for each audio sample is the average score given by the six evaluators.

To measure the consistency of the six raters' scores for each audio sample, inspired by study [4], we calculate the variational ratio and the coefficient of variation for each sample. The variational ratio is a straightforward measure of statistical dispersion for nominal distributions. It is defined as the proportion of observations that do not belong to the most frequently occurring category. The average variational ratio is 0.42 in our dataset, indicating that for each audio sample, approximately 58% of the evaluators tend to give the same score. The coefficient of variation is defined as the ratio of the standard deviation to the mean. It provides a normalized measure of dispersion, reflecting the relative variability within a frequency distribution. The average coefficient of

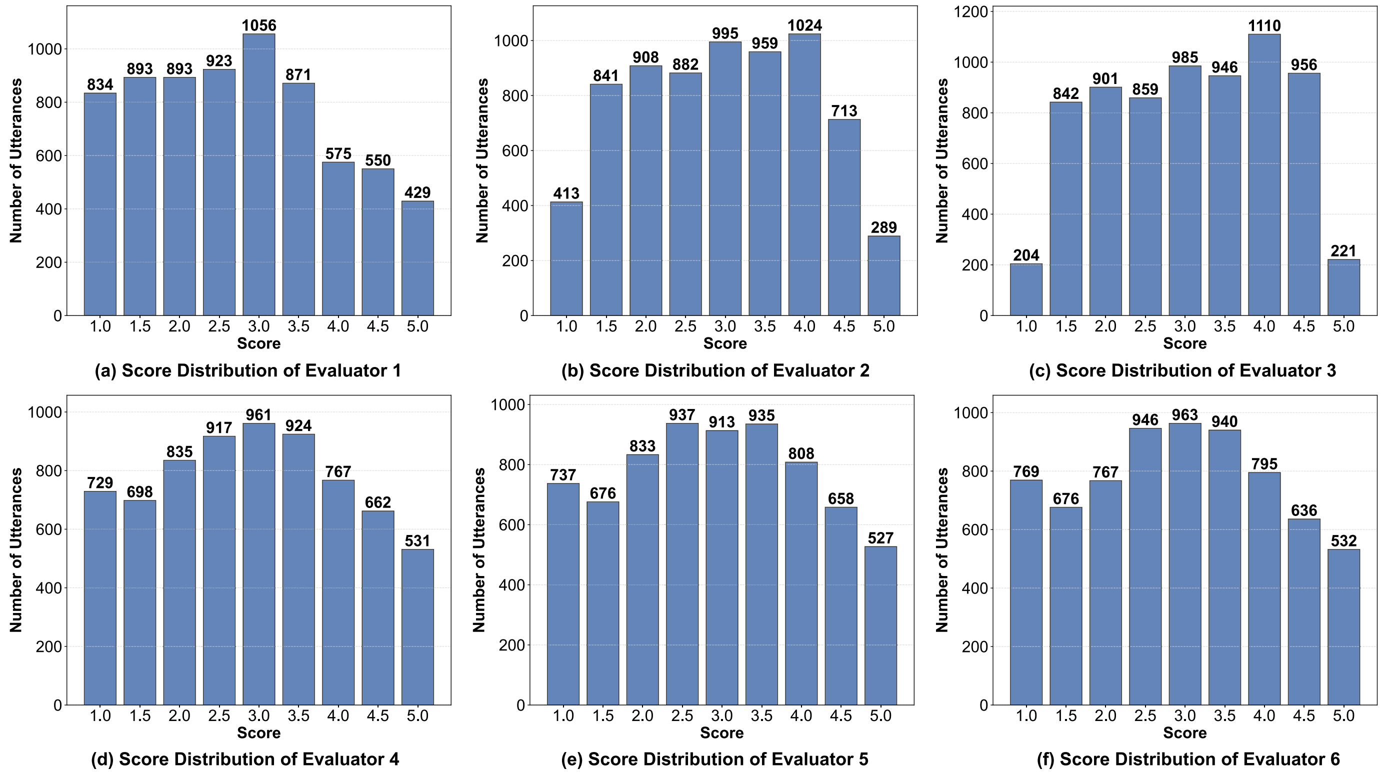

Fig. 1. The score distribution of the six evaluators in GOSMOS dataset.

variation over the dataset is 0.1441, indicating that the scores are relatively stable, with a high degree of consistency among evaluators. This suggests that the dataset demonstrates strong reliability.

3.3. *Dataset composition*

The final dataset consists of 7,024 audio samples across 22 systems, with 5,608 samples for training, 708 for validation, and 708 for testing. This dataset, which we name GOSMOS, is the world's first MOS prediction dataset for Gezi Opera singing voices. To ensure that models trained on GOSMOS can predict both human-performed and machine-synthesized Gezi Opera singing audio, our dataset includes both types of audio. Furthermore, to enable the model to predict across different audio quality levels and singing proficiency, we carefully select samples with a wide range of MOS, ensuring balanced coverage of both high-quality and low-quality audios. All audio files are monophonic with a sampling rate of 16 kHz. The dataset follow the CC BY-NC 4.0 protocol. The dataset can publicly available and only be used for scientific research.

The overall dataset and the data sources of each subset are shown in Fig. 2. In Fig. 2, subfigures (a) through (d) display the data source distributions for the full dataset, training set, validation set, and test set, respectively. In Fig. 2, sys_1 to sys_5 (blue) are recordings of human singers, sys_6 to sys_12 (yellow) are audio generated by SVS models, and sys_13 to sys_22 (green) are low-quality audios created manually based on the first two categories. These three parts of data correspond to the three sources in the data collection. As shown in Fig. 2, the data sources for each category are roughly evenly distributed, ensuring fairness in model training.

In Fig. 3, subfigures (a) through (e) show the MOS distributions of all utterance-level audios, all system-level audios, the training set, the validation set, and the test set in the GOSMOS dataset. We can see that the MOS range from 1 to 5 in each subfigure, with a balanced distribution. As a result, models trained on this dataset are capable of providing reliable prediction scores for various Gezi Opera audios.

4. Method

In this section, section 4.1 introduces the structure and design reasons of our proposed framework. Section 4.2 introduces the MOS prediction module, with a focus on feature processing and loss function. Section 4.3 introduces the integrated feature analysis module.

4.1. *Overview of our framework*

Our MOS prediction framework involves two main components: the MOS prediction module and the integrated feature analysis module. Fig. 4 illustrates the components of our proposed framework.

In the MOS prediction module, our model evaluates singing performance based on five perspectives: initial perception, pronunciation clarity, singer timbre, pitch and emotion. Study [13] indicates that human experts prioritize timbre, text, pitch, and expression when assessing singing. In this study, we treat "text" as the pronunciation clarity and "expression" as the emotion. Inspired by study [13] and considering practical scenarios, we define a combination of five perspectives for our model to predict the MOS. Specifically, we concatenate the five features and feed them into a fully connected layer to predict the MOS. The rationale behind initial perception design is that when we hear a singing voice, we typically form an initial overall subjective impression before evaluating it in more detail from multiple perspectives. The integrated feature analysis module provides feature-level auxiliary textual information to help users understand the singing voice.

4.2. *Singing voice MOS prediction*

For singing voice audio, the most crucial aspects of the MOS prediction task are feature extraction and feature fusion. Due to the outstanding performance of speech SSL models, we adopt the pre-trained wav2vec2.0 speech SSL model as the backbone and utilize several pre-trained speech models to extract features. Fig. 5 illustrates the structure of our proposed MFFMOS model.

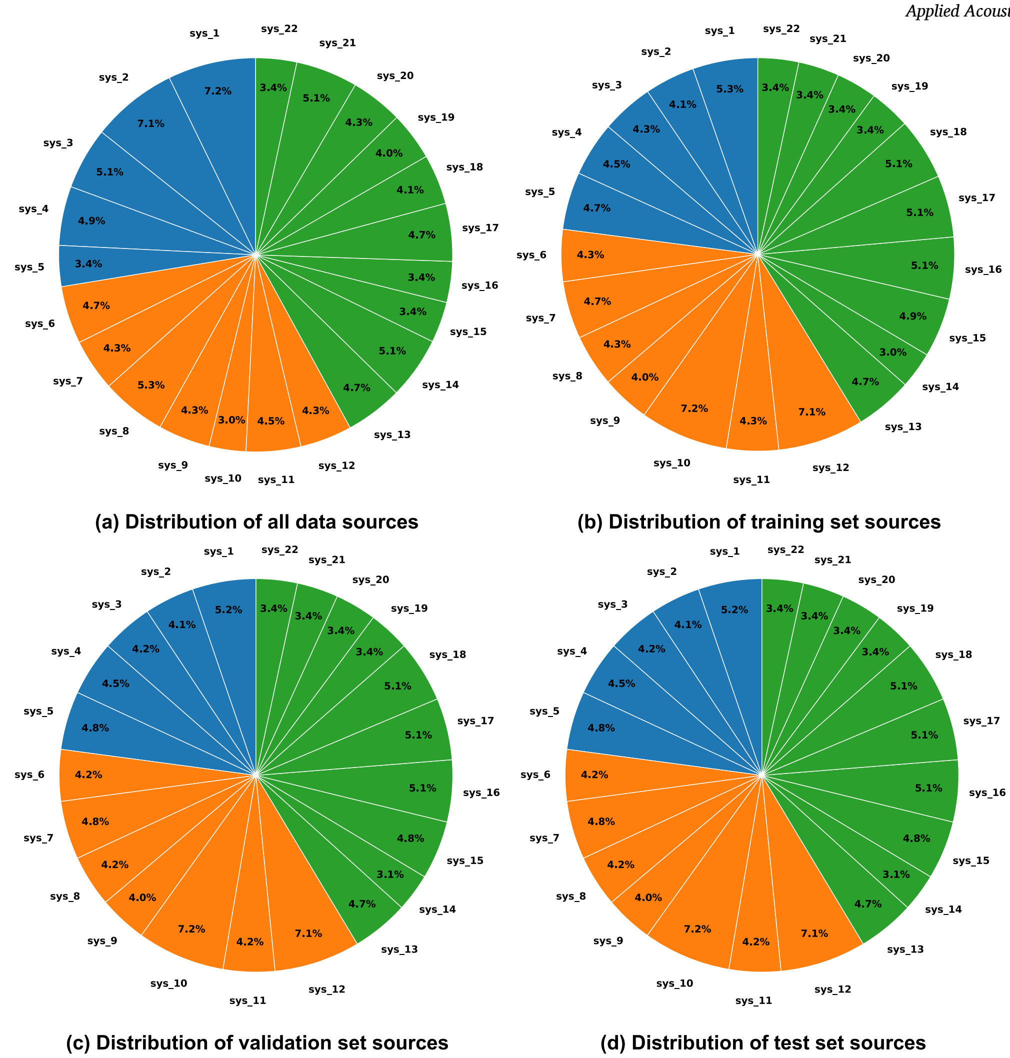

Fig. 2. The data sources distribution of the GOSMOS dataset.

4.2.1. Feature extraction and fusion

Initial perception representation. Wav2vec2.0 model [30] learns speech powerful representation on unlabeled speech data. We choose the wav2vec2.0 base model, whose input is the singing voice to be evaluated and the output is a 768-dimensional vector. We use $\mathbf{v}_o$ to represent the initial perception vector.

Pronunciation clarity representation. We use speech recognition model to evaluate pronunciation clarity. We choose the most powerful speech recognition models today, Massively Multilingual Speech (MMS) model [31] and Whisper [32]. MMS model is released by Meta, which is a speech recognition model that supports over $1,000$ languages and also includes support for Hokkien. Since Gezi Opera is sung in the Hokkien dialect, we can only choose this model. Hokkien representation is a 51-dimensional vector. For singing in other languages, such as English or Chinese, we choose the Whisper_large model. The output representation is a 1280-dimensional vector. We use $\mathbf{v}_c$ to represent the pronunciation clarity vector.

Singer timbre representation. We select the pre-trained speaker verification model [33] released by Google to extract singer timbre representations from singing voice. This model has been trained using a generalized end-to-end loss function. Singer timbre representation is a 256-dimensional vector. We use $\mathbf{v}_s$ to represent the singer timbre vector.

Pitch histogram representation. Previous study [12] has shown that pitch histogram representation is more effective than directly extracting pitch representation in the MOS prediction task for singing voice, so we choose pitch histogram representation. The calculation of the pitch histogram representation is divided into two steps.

The first step is to compute the frequency H of each pitch interval. H_i is the frequency of the i-th pitch interval. The calculation process is shown in Equation (1) and Equation (2):

$$H_i = \sum_{\mathbf{p}_t \in \mathcal{B}_i} 1, \quad i = 1, 2, \dots, B, \tag{1}$$

$$\mathcal{B}_i = \left[\frac{i-1}{B}(f_{max} - f_{min}) + f_{min}, \frac{i}{B}(f_{max} - f_{min}) + f_{min} \right], \quad i = 1, 2, \dots, B, \tag{2}$$

where $\mathcal{B}_i$ is the range of the i-th pitch interval. $B = 120$ is the total number of pitch bins. f_{min} is the minimum pitch, with a value of 30 Hz. f_{max} is the maximum pitch, with a value of 1 kHz. $\mathbf{p}_t$ is the pitch value of the audio at each frame t.

The second step normalizes H_i to obtain the pitch histogram representation $\mathbf{v}_p$, and $\mathbf{v}_p$ is a 120-dimensional vector. The calculation process is shown in Equation (3):

$$\mathbf{v}_p = \frac{H_i}{\sum_{i=1}^{B} H_i}, \quad i = 1, 2, \dots, B. \tag{3}$$

Emotion representation. The model we choose to extract emotion features from singing voice is a strong speech emotion representation model, emotion2vec [34]. The emotion2vec model has proven to per-

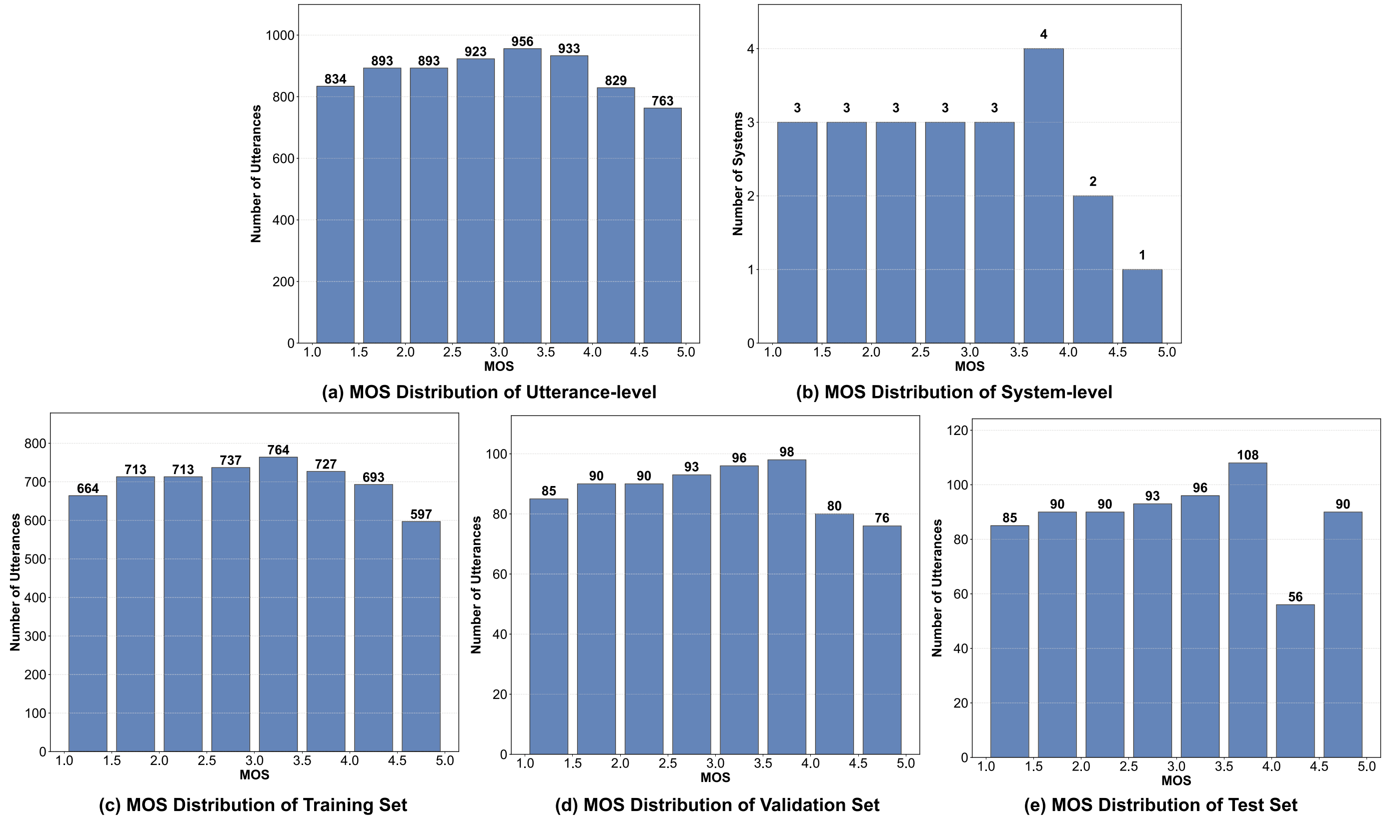

Fig. 3. The MOS distribution in GOSMOS dataset.

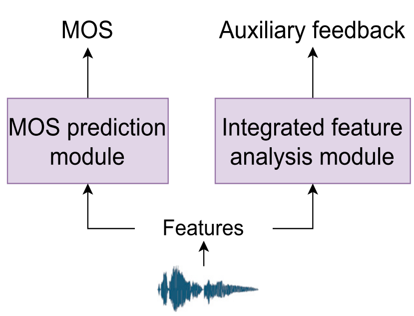

Fig. 4. The components of our proposed framework.

form well in singing emotion recognition task. The input of this model is the singing audio to be evaluated, and the output is an emotion representation vector. We select the emotion2vec base model, which outputs a 768-dimensional vector. We use $\mathbf{v}_e$ to represent the emotion vector.

Feature fusion is also a crucial step, we adopt the concatenation method for fusing feature vector representations. Equation (4) represents the feature concatenation:

$$\mathbf{v}_{concat} = \left[\mathbf{v}_o, \mathbf{v}_c, \mathbf{v}_s, \mathbf{v}_p, \mathbf{v}_e\right], \tag{4}$$

where $\mathbf{v}_{concat}$ is the fused vector representation. The $\mathbf{v}_{concat}$ feature dimension is $1,963$ for Hokkien Gezi Opera and $3,192$ for the other language singing voices.

4.2.2. Training

During the MFFMOS model training process, we freeze the parameters of the pre-trained models for the four features (pronunciation clarity, singer timbre, pitch histogram and emotion) and update only the parameters of the wav2vec2.0 base model. The pre-trained wav2vec2.0 base model serves as the backbone, with its output features concatenated with the other four features before being passed into a single fully connected layer. This approach leverages these four features to fine-tune the wav2vec2.0 base model, enabling effective MOS prediction. In Fig. 5, we also indicate whether the parameters of each pre-trained model are updated. A flame icon represents updated parameters, while a snowflake icon represents frozen parameters. We train a model independently for each dataset.

4.2.3. Loss function

For the singing voice MOS prediction, our loss function is an L1 loss. The formula for L1 loss is shown in Equation (5):

$$L = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right|, \tag{5}$$

where N is the number of sample, y_i is the true MOS for the i-th sample, and $\hat{y}_i$ is the predicted MOS for the i-th sample. $y_i \in [1,5]$ and $\hat{y}_i \in [1,5]$.

4.3. Integrated feature analysis module

At present, all MOS prediction models provide only overall scores, offering limited insights into specific weaknesses, which makes it challenging to improve vocal quality without fine-grained analysis. We design this integrated feature analysis module, which conducts feature-level analysis from four perspectives: pronunciation clarity, timbre, pitch, and emotion, to provide users with supportive feedback for MOS prediction. Fig. 6 is the structure diagram of our integrated feature analysis module.

Pronunciation Clarity. We use the pre-trained speech recognition model to predict the lyrics or phonemes transcription sequence of the singing voice. The Word Error Rate (WER) can only be computed between this sequence and the real sequence of the singing voice. A lower WER indicates better pronunciation clarity of the singing voice.

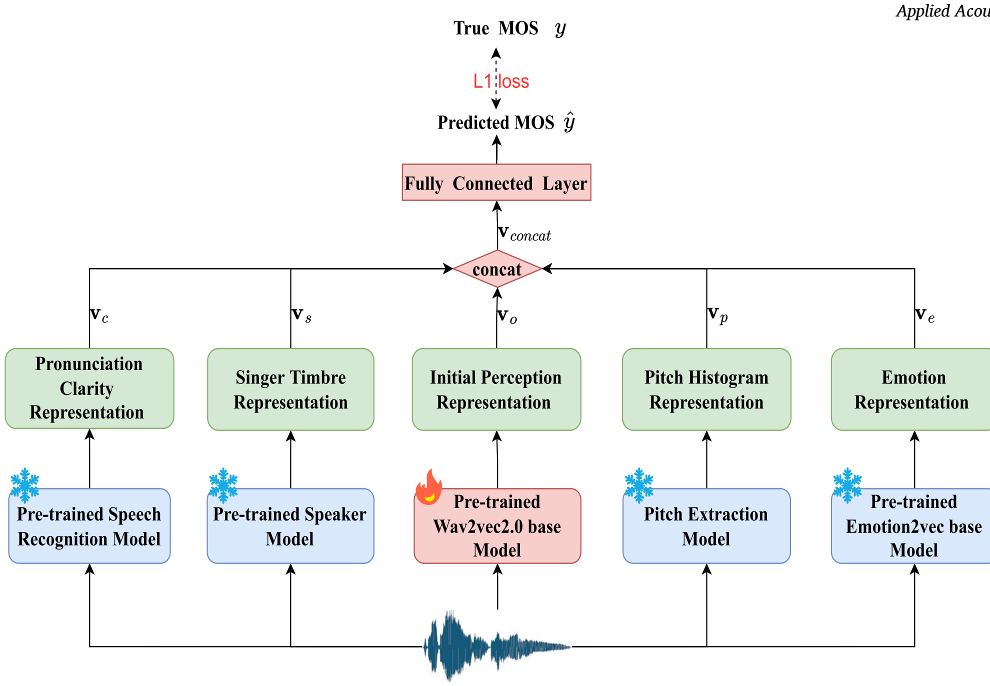

Fig. 5. The structure of our proposed MFFMOS model.

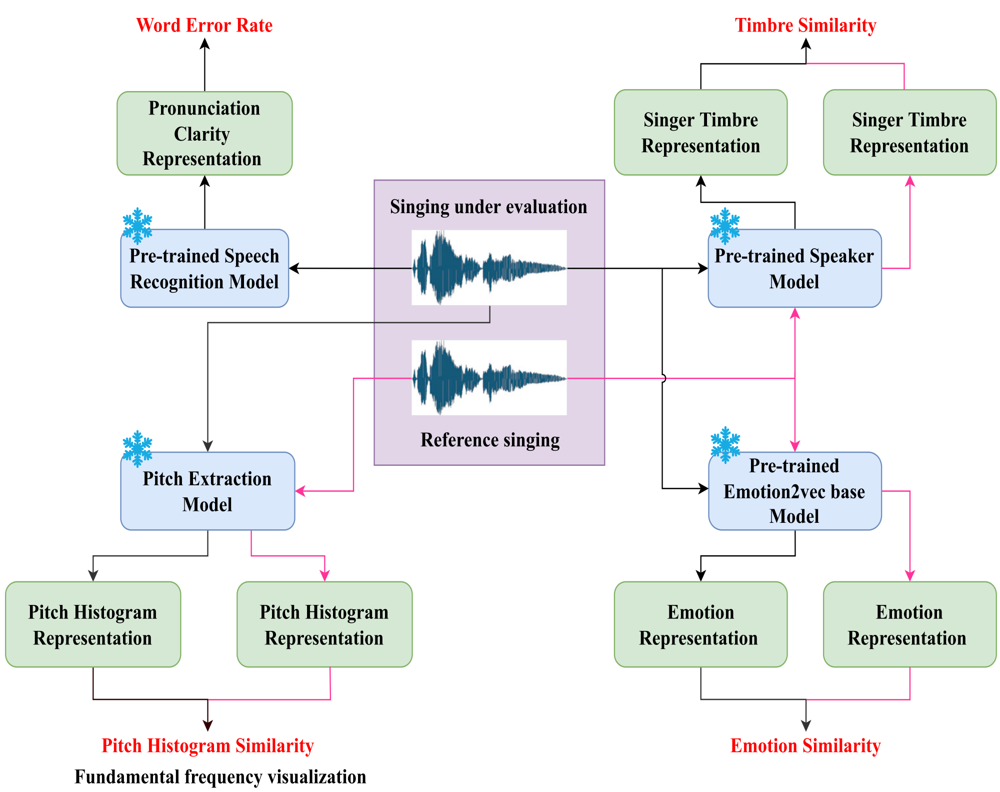

Fig. 6. The structure diagram of our integrated feature analysis module.

Timbre. We use the pre-trained speaker verification model to extract the singer's timbre representation and calculate the cosine similarity of timbre between the singing under evaluation and the original singer's voice. The higher the timbre similarity, the closer the singing voice is to the original singer's timbre.

Pitch. We use a pitch extraction model to extract the pitch and calculate the cosine similarity of pitch histogram representations between the singing under evaluation and the reference singing voice. Since this similarity might be too abstract for users, we visualize the fundamental frequency (F0) curve and provide this feedback.

Emotion. We use a pre-trained emotion2vec base model to extract general emotional representations and calculate the emotional similarity between the singing under evaluation and the reference singing.

Table 1
Comparison of all the datasets we used.

Dataset	Language	Task	Duration (min)	Train	Valid	Test
GOSMOS	Hokkien	Singing	993	5608	708	708
Lyra-SA	Chinese	Singing	3307	12100	4400	3700
VCC2018	English	Speaking	1145	13580	3000	4000
SOMOS_clean	English	Speaking	1570	14100	3000	3000

Table 2
Comparison of pre-trained models during training.

Representation	Pre-trained Model	Feature Dimension	Parameter Update
Initial perception	Wav2vec2.0 base	768	Yes
Singer timbre	Speaker verification	256	No
Pronunciation clarity	MMS(Hokkien)	51	No
Pronunciation clarity	Whisper(Other language)	1280	No
Pitch histogram	-	120	No
Emotion	Emotion2vec base	768	No

5. Experiments

5.1. Datasets

Gezi Opera Singing voice MOS Prediction Dataset. The Gezi Opera singing voice MOS prediction dataset we constructed is detailed in Section 3.

Lyra-Singing Assessment Dataset. Lyra-Singing Assessment (Lyra-SA) dataset [2] is the first public dataset for complete-song singing evaluation released by Tencent Tianqin Music Laboratory in China. The dataset samples are sourced from authorized users of the Wesing app. It comprises 10 songs, each with 100 vocal dry tracks (singing without background music), along with corresponding MIDI and lyric files. The audio is resampled to 44.1 kHz, with a bit depth of 16 bits. Songs 1 to 6 form the training set, songs 7 and 8 form the validation set, and songs 9 and 10 form the test set. The rating range is from 0 to 1, and the evaluators are three singing enthusiasts. Each sample in the original dataset has a duration of 3 to 4 minutes. However, our model or hardware is not capable of handling such long data. Therefore, we use the Praat[3] tool to segment the sample according to the lyrics' sentences, resulting in data samples typically less than 10 seconds. Since the original dataset provide MOS for an entire song, we predict a score by concatenating and processing all the segmented samples from a song.

In addition, we also validate our model on two speech synthesis evaluation datasets.

VCC2018 Dataset. The VCC2018 dataset [35] is a publicly available dataset for evaluating the quality of synthesized speech. The dataset includes 20,580 samples, each rated by 3 to 4 listeners.

Samsung Open Mean Opinion Scores Dataset. The Samsung Open Mean Opinion Scores (SOMOS) dataset[4] is a publicly available dataset for evaluating speech synthesis [36]. The dataset consists of 20,000 synthesized utterances, 100 natural utterances, and 374,955 naturalness evaluations. The naturalness evaluations are scored by several crowd-sourced listeners for the quality of each audio, with ratings ranging from 1 to 5. We actually choose the clean version, not the full version.

We compare the datasets used in this article from four perspectives: language, task, duration and dataset partitioning. The information are shown in Table 1.

5.2. Comparative models

- The **CRNN** [12] model is a convolutional recurrent neural network, specifically designs for the task of singing voice MOS prediction. The input to the model is the CQT spectrogram of the audio.
- The **MOSNet** [22] model is a pioneering work in predicting MOS using neural networks. It is a model established by adopting CNN and RNN layers. The model input is the magnitude spectrogram of the audio.
- The **NISQA** [25] model a MOS prediction model based on CNN, which uses a self-attention network for modeling temporal dependencies. In addition to predicting the overall MOS, the model also predicts four speech quality dimensions: noise, coloration, discontinuity, and loudness. The input to the model is the mel spectrogram of the audio.
- The **LDNet** [24] model is a unified framework for predicting MOS that estimates the perceived quality for each listener based on the input audio and their identity. It adopts an encoder-decoder architecture, with the encoder based on CNN and the decoder based on RNN. The input to the model is the magnitude spectrogram of the audio.
- The **SSL_MOS (SMOS)** [14] model is a fine-tuned model based on speech SSL model (wav2vec2.0 base), which has better generalization ability for zero-shot or out-of-domain speech samples. The input to the model is the audio.
- The **CRNN_PH+SSL (CPS)** [1] model is the model that adds pitch histogram feature and speech SSL model (wav2vec2.0 base) feature to CRNN model. It is a model specifically designed for predicting singing voice MOS. This model achieves the current state-of-the-art (SOTA) performance. The model input is the CQT spectrogram and audio.
- The **MFFMOS** model is our proposed model. The input to the model is the audio.

5.3. Training details

During the training of the MFFMOS model, we use the pre-trained timbre encoder model[5] to extract singer timbre representations, the pre-trained MMS model[6] or pre-trained Whisper_large model[7] to extract the pronunciation clarity representation, and the pre-trained emotion2vec base model[8] to extract the emotion representation. The pitch histogram representation does not utilize pre-trained models and it is simply a standard feature extraction method. During training, the pre-trained wav2vec2.0 base model[9] is only fine-tuned, while other pre-trained models are only used for initial feature extraction. The fully connected layer consists of a single layer, with an input dimension equal to the sum of the feature dimensions from the five aspects and an output dimension of 1. The input dimension is $1,963$ for Hokkien Gezi Opera and $3,192$ for other languages, due to differences in the output dimensions of the speech recognition models used. The pre-trained models and the output feature dimensions are shown in Table 2. The optimizer used for training is SGD with a learning rate of 0.0001, and the loss function is L1 loss. We use an NVIDIA A40 GPU for training, and the best model is saved if the validation loss does not decrease for 20 consecutive epochs.

To ensure a fair comparison, we re-train all comparative models using their official implementations and follow the best hyperparameter configurations as reported in their original papers. Unless otherwise specified, all unspecified parameters follow the default values provided

[2] https://lyracobar.y.qq.com/singvoicedataset.html.
[3] https://github.com/praat/praat.
[4] https://zenodo.org/records/7119400#.ZCorKy8Rr0o.

[5] https://drive.google.com/drive/folders/15oeBYf6Qn1edONkVLXe82MzdIi3O_9m3.
[6] https://github.com/facebookresearch/fairseq/tree/main/examples/mms.
[7] https://github.com/openai/whisper.
[8] https://github.com/ddlBoJack/emotion2vec.
[9] https://github.com/facebookresearch/fairseq/blob/main/examples/wav2vec/README.md.

in the original codebases. The detailed training setups are as follows. MOSNet is trained using the Adam optimizer with a learning rate of 0.0001. The training loss is L1, and the model is trained for 100 epochs with a batch size of 20. NISQA adopts the Adam optimizer with a learning rate of 0.001. The training loss is also L1. We train the model for 100 epochs with a batch size of 40. LDNet uses the RMSProp optimizer with a learning rate of 0.001. The training loss is L1, and the model is trained with a batch size of 30 for a total of 100,000 steps. CRNN is optimized using Adam with a learning rate of 0.0001. The training loss is L1, the batch size is set to 30, and training proceeds for 100 epochs. SMOS and CPS are both trained using the SGD optimizer with a learning rate of 0.0001. The training loss is L1, and both models are trained for 100 epochs with a batch size of 40. Early stopping based on validation loss with a patience of 20 epochs is applied. All models are trained on the same hardware, and evaluation metrics are computed using the same scripts for consistency.

5.4. Evaluation metrics

For the singing voice MOS prediction task, the evaluation metrics are the following four: Mean Square Error (MSE), Linear Correlation Coefficient (LCC), Spearman Rank Correlation Coefficient (SRCC), Kendall Tau Rank Correlation Coefficient (KTAU), and are divided into utterance-level and system-level. The utterance-level refers to the evaluation of individual samples separately, while the system-level means that if several audio samples are generated by the same person or the same SVS model, these samples are considered as belonging to the same system. MSE reflects the degree of difference between the predict score and the real score. LCC is a metric for studying the degree of linear correlation between the predict score and the real score. SRCC is a non-parametric metric for assessing the dependence between the predict score and the real score. KTAU is a rank correlation coefficient. These four evaluation metrics are commonly used in this task. Floating point operations per second (flops) refers to the total number of floating-point operations required for a forward pass of a 1-second audio sample, serving as a measure of the model's computational complexity.

6. Results and discussion

This section primarily presents the experimental results and discusses their implications. Section 6.1 introduces the main experimental results across two singing datasets. Section 6.2 incrementally analyzes the features of singer timbre, pitch, pronunciation clarity, and emotion. Section 6.3 analyzes the relative importance of individual features on the final predicted MOS. Section 6.4 assesses the domain adaptability of our model, demonstrating its effectiveness not only across various singing datasets but also in speaking datasets. Section 6.5 compares three feature fusion methods: direct concatenation, attention-based fusion, and gating-based fusion. Section 6.6 presents the GPU memory usage and computational cost of the model in practical deployment scenarios. Lastly, Section 6.7 provides a user study of our framework in real-world singing education inference scenarios.

6.1. Main result analysis

In this section, we present the results of different comparative models on the singing voice evaluation datasets. Table 3 shows the results on the GOSMOS dataset, and Table 4 shows the results on the Lyra-SA dataset.

As depicted in Table 3, the CRNN model in the first row demonstrates the poorest performance in both utterance-level and system-level MSE, indicating that the network structure combining CNN and RNN is not effective in learning the mapping from speech CQT spectrograms to MOS. The MOSNet model in the second row shows improved performance in both utterance-level and system-level MSE, with a notable 33% reduction in utterance-level MSE. This suggests that its deep CNN architecture is more effective than the CRNN model and better at capturing spectrogram details. Compared to MOSNet, the NISQA model in the third row achieves improvements across all metrics except for utterance-level SRCC and KTAU. Specifically, utterance-level MSE is reduced by 30.6%, while system-level MSE decreases by 34.7%. This indicates that the combination of CNN and self-attention mechanisms in this model is more effective, with the self-attention mechanism playing a significant role. The LDNet model in the fourth row outperforms the NISQA model across all metrics except for system-level KTAU. Specifically, it achieves an 8.2% reduction in utterance-level MSE and a 19.4% reduction in system-level MSE. This suggests that its CNN encoder and RNN decoder architecture is well-designed, and the model effectively incorporates listener information, which significantly contributes to performance improvement.

From the fifth to the seventh row, all models utilize speech SSL models, resulting in a significant improvement in performance. In the fifth row, the SMOS model utilizes the wav2vec2.0 base model. Leveraging its inherent generalization, fine-tuning on the GOSMOS dataset leads to significant performance enhancements. Compared to the best-performing LDNet model based on traditional neural networks, all utterance-level metrics show improvement, while system-level MSE also improves to some extent. Specifically, utterance-level MSE is reduced by 8.5%, and system-level MSE decreases by 3.5%. This demonstrates the strong generalization capability of the pre-trained wav2vec2.0 base model in speech representation. When fine-tuned on domain-specific data, it achieves noticeable performance gains over traditional neural network approaches. The CPS model in the sixth row, which builds upon the features of the CRNN and SMOS models by incorporating pitch histogram features, further improves both utterance-level and system-level MSE. This demonstrates that this feature combination approach is effective.

Compared to CPS, our proposed MFFMOS model in the last row achieves the best performance across all utterance-level metrics, as well as the best system-level MSE and LCC. For utterance-level metrics, MFFMOS reduces MSE by 10.3%, increases LCC by 1.2%, SRCC by 1.4%, and KTAU by 1.6% compared to CPS. For system-level metrics, MFFMOS reduces MSE by 0.98% and increases LCC by 0.16%. These results indicate that our MFFMOS model is effective, and our multi-feature fusion approach can comprehensively assess singing voice. The GOSMOS dataset we have constructed is suitable for singing voice MOS prediction task.

Since the Lyra-SA dataset evaluates entire songs, the results from this dataset only include system-level metrics. As depicted in Table 4, our model achieves the best performance in all metrics. In comparison to the SOTA CPS model designed specifically for singing voice evaluation, the MFFMOS model achieves a 15.6% reduction in MSE, a 51.2% increase in LCC, a 49% increase in SRCC, and a 54.3% increase in KTAU. This result indicate that our model still performs well on the Chinese singing dataset and has good domain adaptability performance.

The experimental results across the two singing voice MOS prediction datasets are consistent. The MFFMOS model achieves the best performance in all utterance-level metrics on the GOSMOS dataset, and in all system-level metrics on the Lyra-SA dataset as well. The MFFMOS model incorporates three additional features—singer timbre, pronunciation clarity, and emotion—compared to the CPS model, which instead includes the CQT spectrogram feature. The consistent best results demonstrate that fine-tuning the wav2vec2.0 base model with a combination of singer timbre, pronunciation clarity, emotion, and pitch histogram features outperforms the combination of CQT spectrogram and pitch histogram features. We analyze that the CQT spectrogram feature is relatively generic, whereas singer timbre, pronunciation clarity, and emotion are key features for singing voice MOS prediction. Their combination is more impactful than the CQT spectrogram feature, significantly enhancing prediction results. Our proposed approach of integrating these features to fine-tune the wav2vec2.0 base model has proven to be effective and reliable.

Table 3
Results on the GOSMOS dataset. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
CRNN	0.6731	0.7037	0.6933	0.5198	0.3924	0.9080	**0.9727**	**0.8909**
MOSNet	0.4508	0.8432	0.8520	0.6723	0.3907	0.8517	0.8713	0.7143
NISQA	0.3128	0.8671	0.8348	0.6618	0.2550	0.8885	0.8950	0.7749
LDNet	0.2872	0.8791	0.8609	0.6899	0.2055	0.9182	0.9051	0.7662
SMOS	0.2627	0.8926	0.8751	0.7170	0.1984	0.9148	0.8938	0.7662
CPS	0.2554	0.8925	0.8729	0.7160	0.1830	0.9179	0.8905	0.7662
MFFMOS	**0.2292**	**0.9036**	**0.8853**	**0.7278**	**0.1812**	**0.9194**	0.9040	0.7922

Table 4
Results on the Lyra-SA dataset. Bold indicates the best result (statistically significant $p < 0.05$).

Model	system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑
CRNN	0.0183	0.2224	0.2206	0.1528
SMOS	0.0180	0.1694	0.1970	0.1433
CPS	0.0147	0.4083	0.4097	0.2893
MFFMOS	**0.0124**	**0.6174**	**0.6106**	**0.4464**

6.2. Incremental feature ablation analysis

We conduct feature ablation experiments where we systematically reduce features from the MFFMOS model to observe the impact of each feature on the results.

The ablation study results are shown in Table 5. The first row presents the results of MFFMOS. The second row shows the results after removing the pronunciation clarity feature, where we can observe a decline in all metrics. Building on the removal of the pronunciation clarity feature, the third row shows the results after removing the pitch histogram feature. Compared to the second row, all metrics deteriorate except for utterance-level KTAU and system-level LCC and SRCC. The fourth row shows the results after removing the singer timbre feature, building on the third row. Both utterance-level and system-level MSE further deteriorate, suggesting the importance of singer timbre. In the last row, we remove the emotion feature, leaving only the initial perception features. We can see that both utterance-level and system-level MSE yield the worst results, which highlights the importance of emotion feature in improving MSE. Through the incremental ablation study of individual features, we observe that each feature contributes to the improvement of the evaluation metrics. It is worth noting that incrementally removing features reduces the input dimensionality to the FC layer, which changes the model's parameter count and may introduce confounding factors in the ablation results. This could weaken the conclusion regarding each feature's individual contribution to overall performance.

6.3. Single-feature ablation analysis

We further evaluate the relative importance of individual features on the predicted MOS. As shown in Table 6, removing pronunciation clarity alone leads to an increase of 0.0128 in utterance-level MSE and 0.0086 in system-level MSE. Removing pitch results in an increase of 0.0174 in utterance-level MSE and 0.0095 in system-level MSE. When singer timbre is removed, the utterance-level MSE and system-level MSE increase by 0.0057 and 0.0081, respectively. Removing emotion leads to increases of 0.0107 and 0.0111 in utterance-level and system-level MSE, respectively. Among all features, pitch contributes the most to utterance-level MSE, while emotion is most critical at the system-level MSE. These findings are consistent with practical observations in real singing performances.

6.4. Model domain adaptability analysis

We train a separate model for each dataset and validate the domain adaptability of our proposed model structure on two English speech MOS prediction datasets.

From Table 7 and Table 8, it can be seen that our MFFMOS model consistently achieves optimal performance across all metrics. This demonstrates the effectiveness and strong domain adaptability of our method, validating its applicability to speech MOS prediction tasks as well.

To provide a more intuitive comparison of the performance of different models, we visualize the system-level MOS of four datasets in scatter plot format. From the Fig. 7, it is shown that our MFFMOS model is closer to the true MOS, outperforming the other models across all four datasets. The system-level MOS distribution of the GOSMOS dataset ranges from 1 to 5, and MFFMOS can accurately predict singing voices of various scores, demonstrating excellent performance.

6.5. Feature fusion method analysis

We compare three feature fusion methods: direct concatenation, attention-based fusion, and gating-based fusion. Table 9, Table 10 and Table 11 present the results on the GOSMOS, VCC2018, and SOMOS_clean datasets, respectively. Among the three, direct concatenation consistently achieves the best results in both utterance-level and system-level MSE, and performs competitively across other metrics. Therefore, we adopt concatenation as our final fusion method. We attribute the inferior performance of attention-based and gating-based methods to their need to project all features into a unified dimension, which either up-projects or compresses certain features. This transformation may introduce noise or remove useful information. In contrast, direct concatenation preserves the original representation and dimensionality of each feature, which contributes to its superior performance.

6.6. Model computational cost

To better understand their practical applicability, we evaluate the computational cost of wav2vec2.0-based models on the GOSMOS dataset. As shown in Table 12, SOMOS, which fine-tunes only the wav2vec2.0 base model, exhibits the lowest GPU memory and flops, but its MSE performance is suboptimal. CPS, which incorporates both CRNN-based and pitch histogram features based on wav2vec2.0, incurs slightly higher GPU memory usage and flops, with a reduction in MSE.

Our proposed MFFMOS model integrates additional features, including pronunciation clarity, speaker timbre, pitch histogram and emotion. It achieves the best MSE results among all models but also requires higher GPU memory and flops, which is a limitation of the method. Specifically, the feature extractor for pronunciation clarity is the most demanding in terms of both GPU memory usage and flops, while the emotion feature extractor contributes significantly to flops. Despite the relatively high GPU memory consumption and flops, this computational overhead is considered a necessary and reasonable trade-off for enhanced performance.

Table 5
Incremental feature ablation results on GOSMOS dataset. **-p** represents removing the pitch histogram feature. **-c** represents removing the pronunciation clarity feature. **-s** represents removing the singer timbre feature. **-e** represents removing the emotion feature. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
MFFMOS	**0.2292**	**0.9036**	**0.8853**	0.7278	**0.1812**	0.9194	**0.9040**	**0.7922**
-c	0.2420	0.9027	0.8760	0.7244	0.1898	0.9171	0.8848	0.7489
-p-c	0.2466	0.8998	0.8852	**0.7303**	0.1877	**0.9195**	**0.9040**	0.7749
-p-c-s	0.2480	0.8951	0.8733	0.7153	0.1910	0.9140	0.8803	0.7403
-p-c-s-e	0.2627	0.8926	0.8751	0.7170	0.1984	0.9148	0.8938	0.7662

Table 6
Single-feature ablation results on GOSMOS dataset. **-p** represents removing the pitch histogram feature. **-c** represents removing the pronunciation clarity feature. **-s** represents removing the singer timbre feature. **-e** represents removing the emotion feature. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
MFFMOS	**0.2292**	0.9036	**0.8853**	**0.7278**	**0.1812**	0.9194	0.9040	0.7922
-c	0.2420	0.9027	0.8760	0.7244	0.1898	0.9171	0.8848	0.7489
-p	0.2466	0.8988	0.8748	0.7158	0.1907	0.9155	0.8893	0.7576
-s	0.2349	**0.9039**	0.8824	0.7254	0.1893	**0.9202**	**0.9209**	**0.8009**
-e	0.2399	0.8986	0.8713	0.7095	0.1923	0.9135	0.8814	0.7489

Table 7
Results on the VCC2018 dataset. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
CRNN	0.5536	0.5612	0.5473	0.3971	0.0949	0.9003	0.9037	0.7411
MOSNet	0.5265	0.5956	0.5580	0.4089	0.1041	0.8976	0.8411	0.6728
NISQA	0.5061	0.6679	0.6392	0.4775	0.0652	0.9684	0.8982	0.7610
LDNet	0.4798	0.6425	0.6068	0.4487	0.0194	0.9879	0.9836	0.9138
SMOS	0.4690	0.7041	0.6776	0.5107	0.0650	0.9721	0.9540	0.8464
CPS	0.4789	0.6920	0.6689	0.5039	0.0229	0.9873	0.9624	0.8634
MFFMOS	**0.3757**	**0.7370**	**0.7108**	**0.5403**	**0.0126**	**0.9906**	**0.9705**	**0.8834**

Table 8
Results on the SOMOS_clean dataset. Bold indicates the best result. Due to the lack of a separate evaluation of individual scores in this dataset, we are unable to report the results of the LDNet model (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
CRNN	0.3235	0.3017	0.3044	0.2075	0.0896	0.4987	0.5314	0.3691
MOSNet	0.3195	0.1507	0.1596	0.1080	0.1097	0.1853	0.2394	0.1637
NISQA	0.5879	0.5469	0.5239	0.3679	0.4023	0.7924	0.7885	0.5858
SMOS	0.2102	0.6204	0.6160	0.4374	0.0328	0.8593	0.8700	0.6771
CPS	0.2094	0.6273	0.6211	0.4429	0.0296	0.8735	0.8833	0.6907
MFFMOS	**0.1768**	**0.6828**	**0.6755**	**0.4882**	**0.0236**	**0.8964**	**0.9042**	**0.7213**

Table 9
Feature fusion method analysis results on the GOSMOS dataset. MFFMOS uses concat fusion; MFFMOS_a and MFFMOS_g use attention-based and gating-based fusion, respectively. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
MFFMOS	**0.2292**	**0.9036**	**0.8853**	**0.7278**	**0.1812**	**0.9194**	0.9040	**0.7922**
MFFMOS_a	0.2596	0.8958	0.8755	0.7274	0.2071	0.9127	0.8848	0.7576
MFFMOS_g	0.2596	0.8919	0.8740	0.7169	0.1819	0.9187	**0.9051**	0.7749

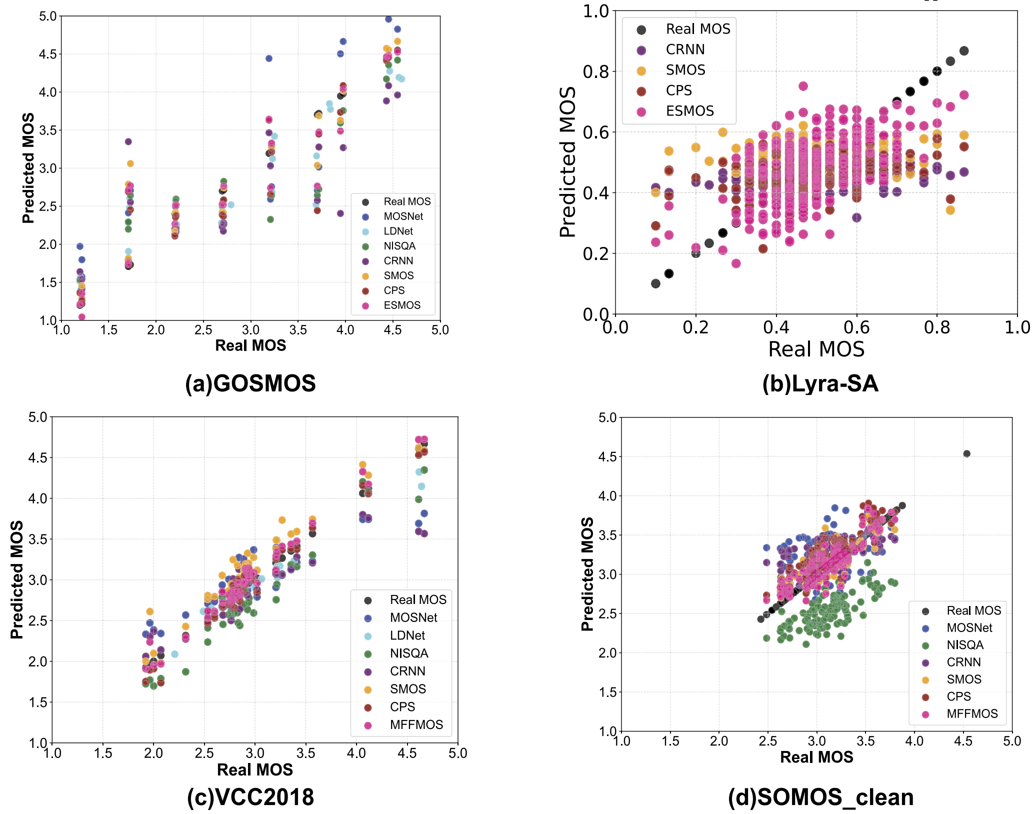

Fig. 7. Scatter plot of system-level results for each dataset.

Table 10
Feature fusion method analysis results on the VCC2018 dataset. MFFMOS uses concat fusion; MFFMOS_a and MFFMOS_g use attention-based and gating-based fusion, respectively. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
MFFMOS	**0.3757**	**0.7370**	**0.7108**	**0.5403**	**0.0126**	**0.9906**	**0.9705**	0.8834
MFFMOS_a	0.4083	0.7250	0.6973	0.5283	0.0211	0.9854	0.9560	0.8663
MFFMOS_g	0.3845	0.7237	0.7005	0.5316	0.0127	0.9879	0.9700	**0.8862**

Table 11
Feature fusion method analysis results on the SOMOS_clean dataset. MFFMOS uses concat fusion; MFFMOS_a and MFFMOS_g use attention-based and gating-based fusion, respectively. Bold indicates the best result (statistically significant $p < 0.05$).

Model	utterance-level				system-level			
	MSE↓	LCC↑	SRCC↑	KTAU↑	MSE↓	LCC↑	SRCC↑	KTAU↑
MFFMOS	**0.1768**	**0.6828**	**0.6755**	**0.4882**	**0.0236**	0.8964	**0.9042**	**0.7213**
MFFMOS_a	0.1975	0.6718	0.6572	0.4738	0.0243	**0.9034**	0.9035	0.7197
MFFMOS_g	0.1874	0.6537	0.6461	0.4615	0.0254	0.8827	0.8920	0.7046

6.7. User study

To evaluate the practical utility of our proposed framework in the context of singing education, particularly the effectiveness of the integrated feature analysis module, we conduct a user study involving 15 students from Hokkien Gezi Opera arts school. All 15 participants are majoring in Gezi Opera singing. Specifically, 5 students are from the lower grades, 5 from the middle grades, and 5 from the upper grades. All participants are fluent in Hokkien.

We preselect five singing clips, and all participants are asked to perform these clips by imitating the corresponding reference recordings. After each performance, our framework provides both a predicted MOS and detailed feedback based on feature analysis. A case of the feedback provided by our framework is shown in Fig. 8. The integrated feature analysis module provides students with predicted phoneme feedback, al-

Table 12
Model computational cost results on the GOSMOS dataset. MSE is the utterance-level MSE.

Model	MSE	GPU Memory(MiB)	Flops(Giga/s)
SMOS	0.2627	1496	5.54
CPS	0.2554	1512	5.71
MFFMOS	0.2292	9134	49.08

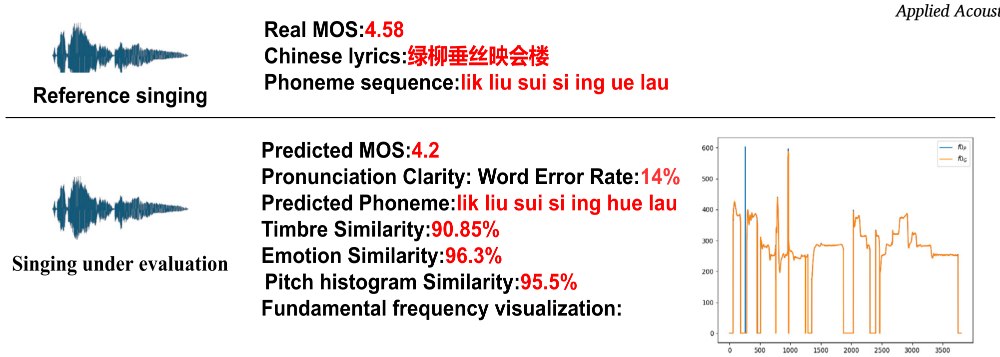

Fig. 8. A case of the feedback provided by our framework.

lowing them to re-evaluate each individual pronunciation. Additionally, by visualizing their fundamental frequency contour, students can assess the accuracy of their singing pitch. This detailed feature-level analysis is essential for helping students promptly identify and address shortcomings in their singing. Once all participants complete the test, we ask them to rate whether the integrated feature analysis module helps them identify specific singing issues and improve their singing performance. Each participant rates their experience on a scale from 1 to 5. Based on the calculation, the average score from our 15 participants is 4.23, with a 95% confidence interval indicating that the true average score is expected to fall between 4.10 and 4.37. This suggests that our integrated feature analysis module is practical in the context of singing education.

7. Conclusion and future work

We construct and publicly release the world's first MOS prediction dataset for Chinese Hokkien Gezi Opera singing, initiating research on Gezi Opera singing voice MOS prediction. We propose a comprehensive framework that consists of a reference-free MOS prediction module and an integrated feature analysis module. The MOS prediction module utilizes a speech self-supervised learning model and incorporates multi-feature fusion to assess singing voice. This is the first model to explicitly include pronunciation clarity and emotion as evaluation features. Meanwhile, the feature analysis module offers reference-based feedback on specific aspects of singing, enabling targeted improvement. Experimental results demonstrate that the MFFMOS model outperforms existing strong methods across key metrics.

Our current research still has certain limitations, and we plan to improve it in the future in the following three areas.

- Parallel training of explainable modules. Currently, the integrated feature analysis module serves as a post-processing step following the MOS prediction module. In future research, we plan to build a dataset that includes evaluations for specific aspects and also expands the duration of the audio samples. In this dataset, evaluators would provide targeted text-based feedback on different aspects of audio quality in addition to giving a score. Such a dataset would support training a model capable of simultaneously predicting MOS and generating explanations, thereby enhancing the model's explainability.
- Adaptation to diverse domains. In the current research, we need to train a separate model for each dataset. We will explore how to fine-tune the model into a universal MOS prediction model that can address challenges posed by data from different singing domains.
- Model Compression. Although speech SSL models have shown performance improvements over traditional small models, their training and inference are resource-consuming. In the future, we aim to explore model compression techniques that reduce model size while maintaining performance, allowing for more efficient resource use.

CRediT authorship contribution statement

Peng Bai: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Formal analysis, Data curation. **Yue Zhou:** Supervision. **Ke Gu:** Resources. **Meizhen Zheng:** Data curation. **Linshujie Zheng:** Validation. **Yidong Chen:** Supervision. **Xiaodong Shi:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis.

Funding

This work is supported by the Major Scientific Research Project of the State Language Commission in the 13th Five-Year Plan (Grant No. WT135-38), and the public technology service platform project of Xiamen City (No. 3502Z20231043).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Special thanks to the actors and actresses who participated in the audio recording of Gezi Opera. Thanks to the Gezi Opera professionals who have rated their scores. Thanks to the students who participated in the user study and provided valuable feedback. We thank the reviewers for their valuable comments.

Data availability

The Gezi Opera MOS prediction datasets we constructed are available for research purposes only and can be accessed with authorization by contacting us. The Lyra-SA dataset is also available upon request for authorized use by contacting Tencent Tianqin Music Laboratory, and we can provide the partitioned data. The VCC2018 dataset and SOMOS_clean dataset are both publicly accessible.

References

[1] Chan Ping-Chen, Chen Po-Wei, Soo Von-Wun. Improve singing quality prediction using self-supervised transfer learning and human perception feedback. In: Cheng Wen-Huang, Chu Wei-Ta, Hu Min-Chun, Liu Jiaying, Kim Munchurl, Zhang Wei, editors. ACM multimedia Asia 2023. Tainan, Taiwan: ACM; 2023:69.

[2] Bai Peng, Zhou Yue, Zheng Meizhen, Sun Wujin, Shi Xiaodong. Improving Chinese pop song and hokkien gezi opera singing voice synthesis by enhancing local modeling. In: Bouamor Houda, Pino Juan, Bali Kalika, editors. Proceedings of the 2023 conference on empirical methods in natural language processing. Singapore: Association for Computational Linguistics; 2023. p. 3302–12.

[3] Zheng Meizhen, Bai Peng, Shi Xiaodong, Zhou Xun, Yan Yiting. FT-GAN: fine-grained tune modeling for Chinese opera synthesis. In: Wooldridge Michael J, Dy Jennifer G, Natarajan Sriraam, editors. Thirty-eighth AAAI conference on artificial intelligence, AAAI 2024, thirty-sixth conference on innovative applications of artificial

intelligence, IAAI 2024, fourteenth symposium on educational advances in artificial intelligence. Vancouver, Canada: AAAI Press; 2024. p. 19697–705.

[4] Zhang Ning, Jiang Tao, Deng Feng, Li Yan. Automatic singing evaluation without reference melody using bi-dense neural network. In: IEEE international conference on acoustics, speech and signal processing. Brighton, United Kingdom: IEEE; 2019. p. 466–70.

[5] Sun Xiaoheng, Gao Yuejie, Lin Hanyao, Liu Huaping. Tg-critic: a timbre-guided model for reference-independent singing evaluation. In: IEEE international conference on acoustics, speech and signal processing ICASSP 2023. Rhodes Island, Greece: IEEE; 2023. p. 1–5.

[6] Ju Yaolong, Xu Chunyang, Guo Yichen, Li Jinhu, Lui Simon. Improving automatic singing skill evaluation with timbral features, attention, and singing voice separation. In: IEEE international conference on multimedia and expo. Brisbane, Australia: IEEE; 2023. p. 612–7.

[7] Tomoyasu Nakano, Goto Masataka, Hiraga Yuzuru. Subjective evaluation of common singing skills using the rank ordering method. In: Ninth international conference on music perception and cognition; 2006. p. 1507–12.

[8] Tan Terry. Singing evaluation based on deep metric learning. In: ISCSIC 2019: 3rd international symposium on computer science and intelligent control. Amsterdam, the Netherlands: ACM; 2019:17.

[9] Nakano Tomoyasu, Goto Masataka, Hiraga Yuzuru. An automatic singing skill evaluation method for unknown melodies using pitch interval accuracy and vibrato features. In: INTERSPEECH 2006 - ICSLP, ninth international conference on spoken language processing. Pittsburgh, PA, USA: ISCA; 2006.

[10] Gupta Chitralekha, Li Haizhou, Wang Ye. Automatic leaderboard: evaluation of singing quality without a standard reference. IEEE/ACM Trans Audio Speech Lang Process 2020;28:13–26. https://doi.org/10.1109/TASLP.2019.2947737.

[11] Tsai Wei-Ho, Lee Hsin-Chieh. Automatic evaluation of karaoke singing based on pitch, volume, and rhythm features. IEEE Trans Speech Audio Process 2012;20(4):1233–43. https://doi.org/10.1109/TASL.2011.2174224.

[12] Huang Lin, Gupta Chitralekha, Li Haizhou. Spectral features and pitch histogram for automatic singing quality evaluation with CRNN. In: Asia-Pacific signal and information processing association annual summit and conference. Auckland, New Zealand: IEEE; 2020. p. 492–9. https://ieeexplore.ieee.org/document/9306350.

[13] Hausknecht Josipa Bainac, Murdaugh Kristen M, Nagl Elke, Herbst Christian T. Global inventory and similarity rating of singing voice assessment terms used at English speaking academic institutions. J Voice 2021.

[14] Cooper Erica, Huang Wen-Chin, Toda Tomoki, Yamagishi Junichi. Generalization ability of MOS prediction networks. In: IEEE international conference on acoustics, speech and signal processing, virtual and Singapore. Virtual and Singapore: IEEE; 2022. p. 8442–6.

[15] Molina Emilio, Barbancho Isabel, Gómez Emilia, Barbancho Ana M, Tardón Lorenzo J. Fundamental frequency alignment vs. note-based melodic similarity for singing voice assessment. In: IEEE international conference on acoustics, speech and signal processing. Vancouver, BC, Canada: IEEE; 2013. p. 744–8.

[16] Gupta Chitralekha, Li Haizhou, Wang Ye. Perceptual evaluation of singing quality. In: 2017 Asia-Pacific signal and information processing association annual summit and conference. Kuala Lumpur, Malaysia: IEEE; 2017. p. 577–86.

[17] Jaiswal Rahul Kumar, Dubey Rajesh Kumar. Caqoe: a novel no-reference context-aware speech quality prediction metric. ACM Trans Multimed Comput Commun Appl 2023;35(1s):35. https://doi.org/10.1145/3529394.

[18] Gamper Hannes, Reddy Chandan KA, Cutler Ross, Tashev Ivan J, Gehrke Johannes. Intrusive and non-intrusive perceptual speech quality assessment using a convolutional neural network. In: 2019 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA); 2019. p. 85–9.

[19] Rix Antony W, Beerends John G, Hollier Michael P, Hekstra Andries P. Perceptual evaluation of speech quality (pesq)-a new method for speech quality assessment of telephone networks and codecs. In: IEEE international conference on acoustics, speech, and signal processing. Salt Lake City, Utah, USA: IEEE; 2001. p. 749–52.

[20] Hines Andrew, Skoglund Jan, Kokaram Anil C, Harte Naomi. Visqol: an objective speech quality model. EURASIP J Audio Speech Music Process 2015;2015:13. https://doi.org/10.1186/S13636-015-0054-9.

[21] Chinen Michael, Lim Felicia SC, Skoglund Jan, Gureev Nikita, O'Gorman Feargus, Hines Andrew. Visqol v3: an open source production ready objective speech and audio metric. In: Twelfth international conference on quality of multimedia experience. Athlone, Ireland: IEEE; 2020. p. 1–6.

[22] Lo Chen-Chou, Fu Szu-Wei, Huang Wen-Chin, Wang Xin, Yamagishi Junichi, Tsao Yu, et al. Mosnet: deep learning-based objective assessment for voice conversion. In: Kubin Gernot, Kacic Zdravko, editors. 20th annual conference of the international speech communication association. Graz, Austria: ISCA; 2019. p. 1541–5.

[23] Leng Yichong, Tan Xu, Zhao Sheng, Soong Frank K, Li Xiang-Yang, Qin Tao. MBNET: MOS prediction for synthesized speech with mean-bias network. In: IEEE international conference on acoustics, speech and signal processing. Toronto, ON, Canada: IEEE; 2021. p. 391–5.

[24] Huang Wen-Chin, Cooper Erica, Yamagishi Junichi, Toda Tomoki. Ldnet: unified listener dependent modeling in MOS prediction for synthetic speech. In: IEEE international conference on acoustics, speech and signal processing. Virtual and Singapore: IEEE; 2022. p. 896–900.

[25] Mittag Gabriel, Naderi Babak, Chehadi Assmaa, Möller Sebastian. NISQA: a deep cnn-self-attention model for multidimensional speech quality prediction with crowdsourced datasets. In: Hermansky Hynek, Cernocký Honza, Burget Lukás, Lamel Lori, Scharenborg Odette, Motlícek Petr, editors. 22nd annual conference of the international speech communication association. ISCA; 2021. p. 2127–31.

[26] Shen Kailai, Yan Diqun, Dong Li. Msqat: a multi-dimension non-intrusive speech quality assessment transformer utilizing self-supervised representations. Appl Acoust 2023;212:109584. https://doi.org/10.1016/j.apacoust.2023.109584.

[27] Qi Zili, Hu Xinhui, Zhou Wangjin, Li Sheng, Wu Hao, Lu Jian, et al. LE-SSL-MOS: self-supervised learning MOS prediction with listener enhancement. In: IEEE automatic speech recognition and understanding workshop. Italy: IEEE; 2023. p. 1–6.

[28] Ren Yi, Hu Chenxu, Tan Xu, Qin Tao, Zhao Sheng, Zhao Zhou, et al. Fastspeech 2: fast and high-quality end-to-end text to speech. In: 9th international conference on learning representations, virtual; 2021. OpenReview.net. https://openreview.net/forum?id=piLPYqxtWuA.

[29] Liu Jinglin, Li Chengxi, Ren Yi, Chen Feiyang, Zhao Zhou. DiffSinger: singing voice synthesis via shallow diffusion mechanism. In: Thirty-sixth AAAI conference on artificial intelligence, AAAI 2022, thirty-fourth conference on innovative applications of artificial intelligence, IAAI 2022, the twelveth symposium on educational advances in artificial intelligence, virtual. AAAI Press; 2022. p. 11020–8.

[30] Baevski Alexei, Zhou Yuhao, Mohamed Abdelrahman, Auli Michael. Wav2vec 2.0: a framework for self-supervised learning of speech representations. In: Larochelle Hugo, Ranzato Marc'Aurelio, Hadsell Raia, Balcan Maria-Florina, Lin Hsuan-Tien, editors. Advances in neural information processing systems 33: annual conference on neural information processing systems 2020, virtual; 2020.

[31] Pratap Vineel, Tjandra Andros, Shi Bowen, Tomasello Paden, Babu Arun, Kundu Sayani, et al. Scaling speech technology to 1, 000+ languages. CoRR. arXiv: 2305.13516 [abs], 2023. https://doi.org/10.48550/ARXIV.2305.13516.

[32] Radford Alec, Kim Jong Wook, Xu Tao, Brockman Greg, McLeavey Christine, Sutskever Ilya. Robust speech recognition via large-scale weak supervision. In: Krause Andreas, Brunskill Emma, Cho Kyunghyun, Engelhardt Barbara, Sabato Sivan, Scarlett Jonathan, editors. International conference on machine learning, proceedings of machine learning research. Proceedings of machine learning research, vol. 202. Honolulu, Hawaii, USA: PMLR; 2023. p. 28492–518. https://proceedings.mlr.press/v202/radford23a.html.

[33] Wan Li, Wang Quan, Papir Alan, López-Moreno Ignacio. Generalized end-to-end loss for speaker verification. In: 2018 IEEE international conference on acoustics, speech and signal processing. Calgary, AB, Canada: IEEE; 2018. p. 4879–83.

[34] Ma Ziyang, Zheng Zhisheng, Ye Jiaxin, Li Jinchao, Gao Zhifu, Zhang Shiliang, et al. emotion2vec: self-supervised pre-training for speech emotion representation. In: Ku Lun-Wei, Martins Andre, Srikumar Vivek, editors. Findings of the association for computational linguistics. Bangkok, Thailand and Virtual Meeting: Association for Computational Linguistics; 2024. p. 15747–60.

[35] Lorenzo-Trueba Jaime, Yamagishi Junichi, Toda Tomoki, Saito Daisuke, Villavicencio Fernando, Kinnunen Tomi, et al. The voice conversion challenge 2018: promoting development of parallel and nonparallel methods. In: Larcher Anthony, Bonastre Jean-François, editors. Odyssey 2018: the speaker and language recognition workshop. Les Sables d'Olonne, France: ISCA; 2018. p. 195–202.

[36] Maniati Georgia, Vioni Alexandra, Ellinas Nikolaos, Nikitaras Karolos, Klapsas Konstantinos, Sung June Sig, et al. SOMOS: the Samsung open MOS dataset for the evaluation of neural text-to-speech synthesis. In: Proc. Interspeech 2022. Korea: Incheon Songdo ConvensiA; 2022. p. 2388–92.